

Supplemental Material for: Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

Xingqiang Chen and Qixing Wang
Department of Physics, Xiamen University, Xiamen 361005, China

I. ARCHIVED LEGACY-PBE TETRAMER BASELINE

Table I reproduces the archived rigid-strain grid (PBE without D3) retained for reproducibility; it is *not* used for quantitative claims in the main text.

II. PERIODIC SYNERGY ROWS ($n \geq 6$)

III. MARCUS REORGANIZATION ENERGIES

Polaron transport diagnostics are orthogonal to the total-energy synergy audit $\mathcal{S}$ in the main text.

IV. HOPPING INTEGRAL J

On the tetramer dimer grid, B substitution at $\epsilon=0$ delocalizes the charged-state orbital (IPR $75.0 \rightarrow 24.6$) while raising J from 26.6 to 46.4 meV; coupled B @ +3% gives $J = 37.2$ meV and IPR 23.1. The criterion $J > \lambda/2$ for polaron hopping dominance [1] is *not* met at PBE+D3 ($J < \lambda/2$), so a polaron transition is not confirmed on this grid.

[1] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa,

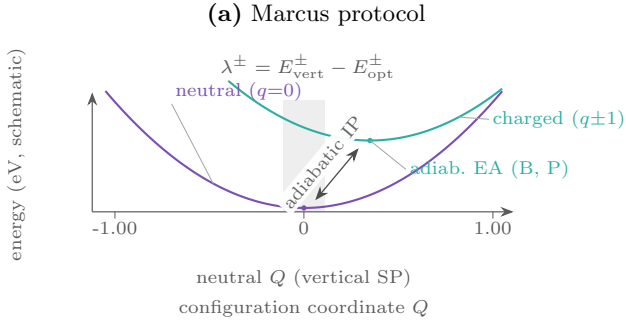
electron localization and mobility in monolayer fullerene networks, Nano Lett. **24**, 7891 (2024).

TABLE I. **Archived baseline only (Supplemental Material).** Linear strain sensitivities and raw substitution energy differences for tetramer models (rigid-strain grid, PBE+D3; reference placement, seed 42). Tabulated ΔE_{sub} values are *not* thermodynamic formation enthalpies (no chemical potentials); use only for relative ranking within this protocol. Main-text quantitative claims use PBE+D3 throughout (Table ?? alternate placement; periodic Table ??). This legacy-PBE grid is retained for reproducibility only. $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon=0$ with no chemical-potential reference (not a standard formation enthalpy). n_{dopants} is counted from archived input coordinates (nominal 5% target; B: 13, N: 12, P: 12). **Not a formation enthalpy:** E_{sub} is a fixed-protocol total-energy difference per dopant site with no elemental chemical-potential reference; use only for *relative* ranking within this study, not standard thermodynamic formation energies.

System	n_{dop}	$E(\epsilon=0)$ (E_h)	$E(\epsilon=+5\%)$ (E_h)	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	−1363.5512	−1363.5509	−1.0	—
B-doped	13	−1328.4900	−1328.4090	+56.9	+73.4
N-doped	12	−1414.3812	−1414.4053	−303.2	−115.3
P-doped	12	−1370.4965	−1370.4888	−19.1	−15.8

TABLE II. Supplemental periodic synergy rows at +3% ($n \geq 6$; 350 Ry production cutoff; completeness only).

n	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
6	B	+3	−7.76
6	N	+3	+2.38
6	P	+3	−31.89
8	B	+3	−11.26
8	N	+3	−2.30
8	P	+3	−29.94



(b) Adiabatic IP/EA and $\lambda^{\pm}$

Dopant	IP (eV)	EA (eV)	λ^{+} (eV)	λ^{-} (eV)
Pristine	4.73	2.61	0.041	0.042
N	3.89	2.77	0.058	0.065
B	4.11	3.12	0.051	— [†]
P	4.07	2.96	0.200	0.315

[†]B channel: $\lambda^{-} < 0$ under fixed neutral-geometry vertical SP; not reported.

FIG. 1. Marcus reorganization energies from charged-state geometry optimizations and vertical single-points at the neutral geometry. Panel (a) schematically defines the vertical-adiabatic decomposition following Ref. 1; panel (b) tabulates adiabatic IP/EA and $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ for pristine and B/N/P tetramers. The electron-channel λ^{-} for B is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

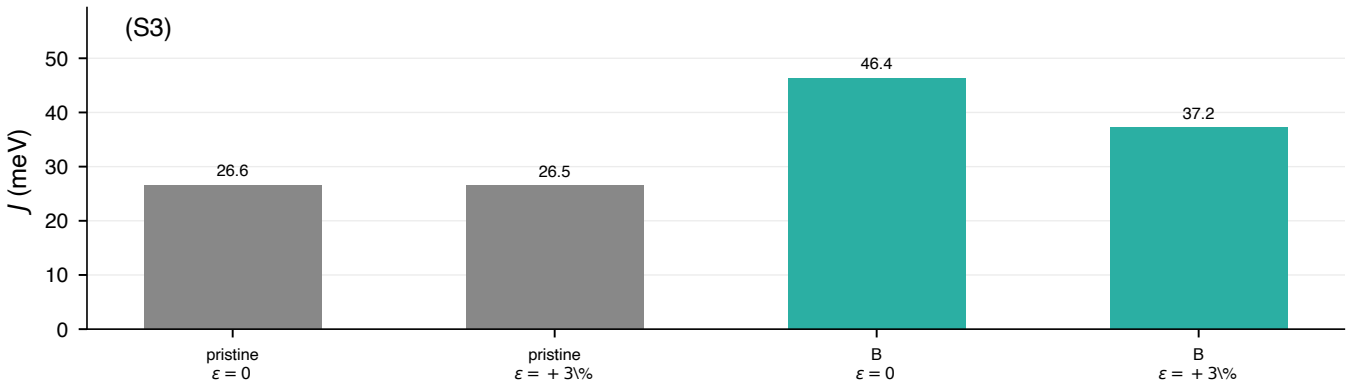


FIG. 2. Nearest-neighbor electronic coupling J on the tetramer dimer grid (pristine and B-doped cells at $\epsilon=0$ and +3%; PBE+D3). Companion inverse-participation-ratio values for the same four corners are tabulated in the open data release (Methods).